

DISCRETE EVENT SIMULATION AND OPTIMIZATION OF A SMARTPHONE TECHNICAL SERVICE CENTER

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EXECUTIVE SUMMARY

This comprehensive study was conducted to identify operational bottlenecks and optimize the service capacity within a high-traffic smartphone technical service center. Initial observations and baseline modeling of the current system revealed severe congestion during the technical repair phase. Specifically, technical personnel were operating at an unsustainable 99.58% utilization rate, pushing the system beyond its practical capacity. Consequently, the average system flow time encompassing both waiting and processing periods escalated to an excessive 143.76 hours per device.

Current Scenario: Continuing operations with the existing resource constraints (2 technical staff members) demonstrated no reduction in service times, while customer attrition (rejected repair quotations) continued at a high rate.

Pilot Scenario: An experimental design was tested wherein technical staff capacity was increased to 3 personnel, and the cashier's idle time was strategically reallocated to include active marketing and customer information tasks.

Decision and Recommendation: According to the simulation outputs, the targeted capacity expansion reduced the total flow time to 83.02 hours, achieving an approximate 42% operational improvement. Furthermore, integrating the cashier into the customer briefing process successfully increased the acceptance rates for paid repairs, thereby enhancing the financial efficiency of the system. Considering both customer satisfaction metrics and operational velocity, the implementation of the Pilot Scenario is strongly recommended.

1. PROBLEM DEFINITION

System Definition and Scope

In this study, the customer traffic and sequential repair stages within a mobile technical service center were meticulously simulated. Customers entering the system at stochastic intervals primarily undergo a warranty verification stage. The process logic realistically models the decision-making behavior of out-of-warranty device owners regarding whether to accept paid diagnostic services, as well as the identification of 'user-induced faults' following the diagnostic phase. In the final stage, customers who approve the presented repair quotation are routed to the repair process, while those who decline exit the system, thereby completing the operational cycle.

Project Objectives

Through this simulation modeling effort, the primary objective is to accurately measure customer density and queuing delays within the service center. By analyzing personnel utilization rates, the operational efficiency of the existing service capacity is critically evaluated. Additionally, the study aims to quantify service loss by tracking the proportion of customers who abandon the system due to expired warranties. The ultimate goal of this project is to generate reliable forecasts regarding the center's daily and weekly revenue potential based on established pricing policies and optimized workflow dynamics.

2. MODEL FORMULATION

System Characteristics and Assumptions

- **Entity Definition:** Entities entering the system are defined as "DEFECTIVE PHONE", and an "ARRIVAL TIME" attribute is assigned to each entity upon entry to facilitate accurate flow time measurement.
- **Queue Structure:** A strict FIFO queuing discipline has been adopted across all processing modules to ensure chronological fairness.
- **Time Unit:** The operational processes of the model are simulated on a "Minute" basis.
- **Interarrival Time:** Statistical goodness-of-fit analysis determined that customer arrival patterns conform to an Exponential distribution.

Service Times

- **Fault Detection:** Updated to an Exponential distribution, EXPO(35) minutes, in the pilot model.
- **Repair Quotation/Pricing:** Modeled uniformly, UNIF(5, 15) minutes.
- **Technical Repair:** Modeled using a Triangular distribution, TRIA (10, 110, 480) minutes, to account for best, most likely, and worst-case repair scenarios.
- **Cashier/Checkout:** Modeled uniformly, UNIF(10, 25) minutes.

Decision Mechanisms Warranty Status: 64% of devices are In-Warranty; 36% are Out-of-Warranty.

- **User-Induced Fault:** Following diagnosis, 22% of device issues are classified as user errors, thereby voiding the warranty.
- **Customer Approval:** The acceptance rate for quoted paid repairs is structurally modeled at 88%.

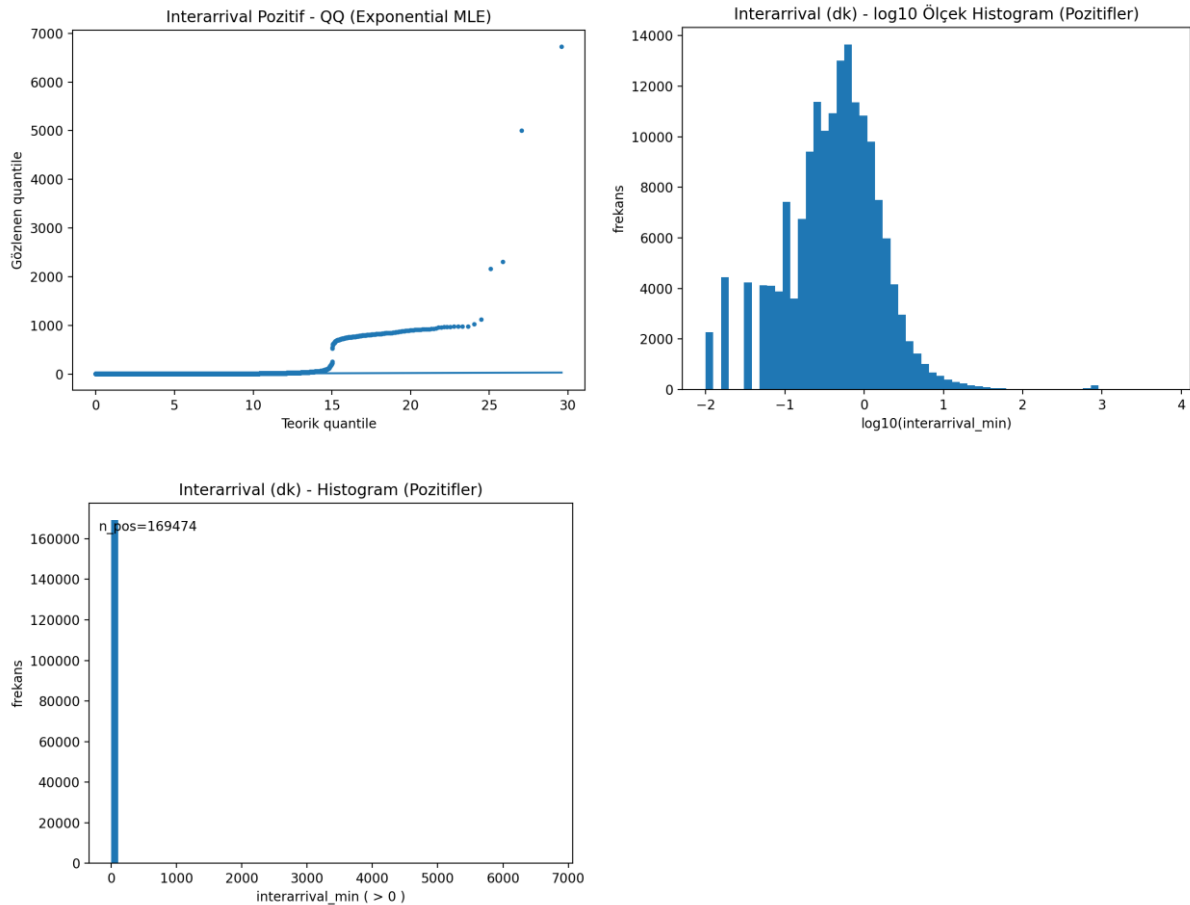
Performance Variables

- **Flow Time:** The total duration a mobile device spends within the system boundaries (Lead Time).
- **Resource Utilization:** The working efficiency and active engagement percentage of technical and administrative personnel.
- **Throughput:** The total volume of successfully repaired devices exiting the system over a defined period.

3. DATA COLLECTION AND ANALYSIS

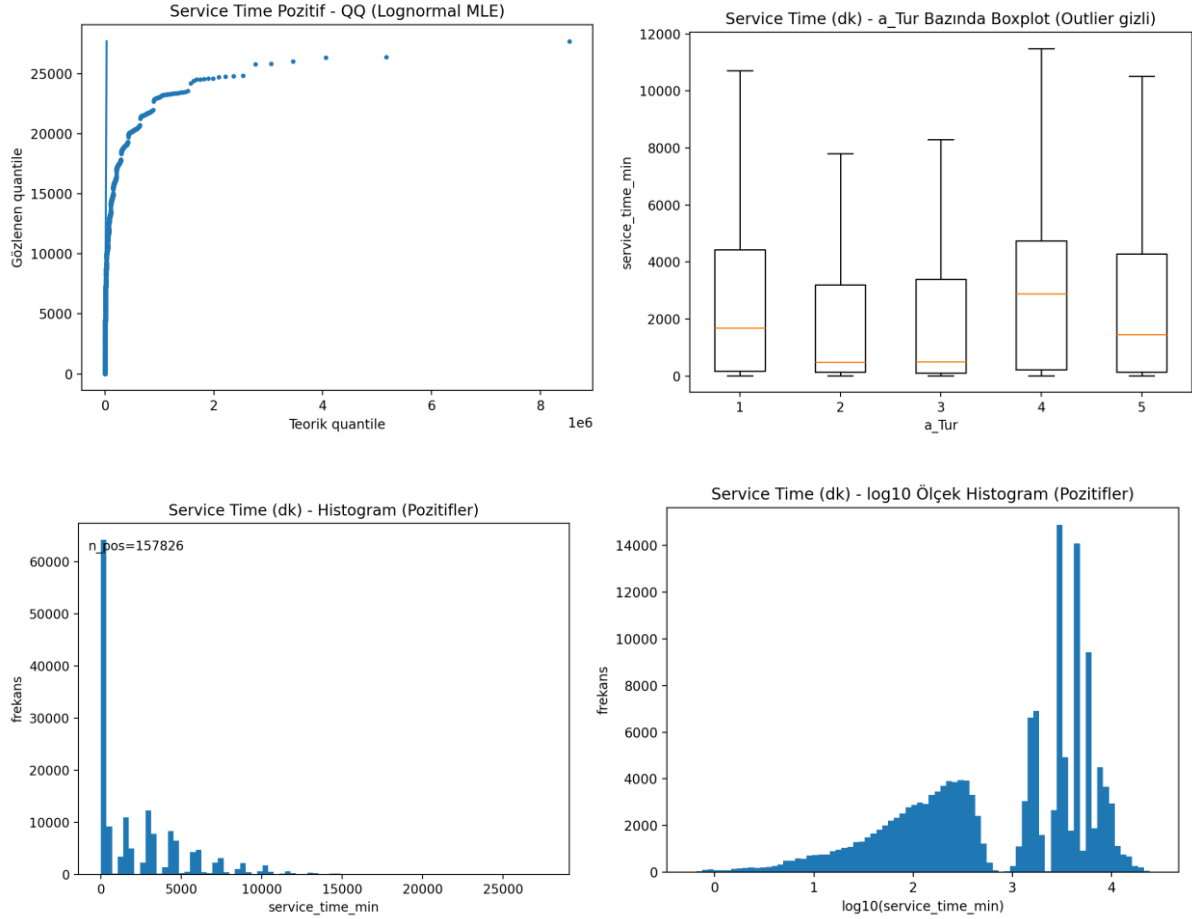
The interarrival and service time variables, which form the foundational inputs of the simulation model, were analyzed using advanced statistical methodologies. During the analysis, only positive and physically meaningful observations from the raw historical dataset were considered.

When analyzing the histograms for the interarrival time variable, it was observed that the data heavily concentrated at lower values, exhibiting a distinct right-skewed structural pattern. Evaluations conducted via Q-Q (Quantile-Quantile) plots revealed a strong linear correlation between the theoretical and observed values; consequently, it was deemed statistically appropriate to represent the interarrival times using an Exponential distribution.



Conversely, the service time data demonstrated a pronounced long-tail (heavy-tailed) behavior, which is typical in complex technical repair environments. Q-Q plots prepared under the assumption of a Lognormal distribution highlighted a particularly strong fit, especially within the mid-range segments of the data. Furthermore, a boxplot analysis conducted to determine the

impact of different fault typologies on service times revealed significant variances in central tendencies and data dispersion. This critical finding justified the necessity of defining distinct service time distributions for varying fault types within the model. Ultimately, the model inputs were established on an empirically driven and statistically robust framework.

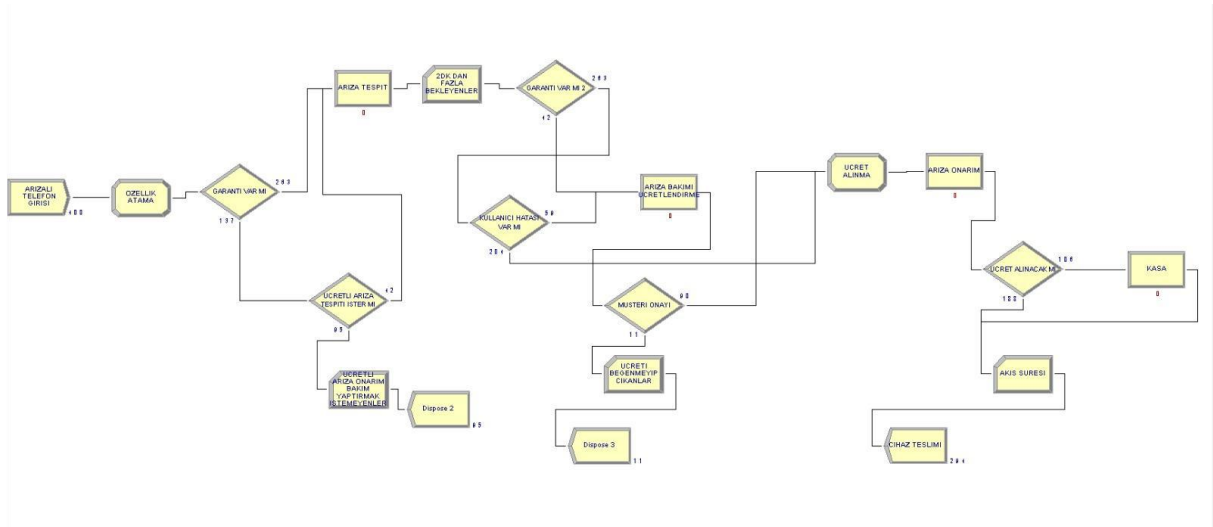


4. COMPUTER MODEL

In this phase of the project, the conceptualized technical service system was translated into a computational model within the ARENA simulation environment, effectively digitalizing the operational logic.

General Flowchart and Modules of the Model

The system initiates with the "Defective Phone Entry" module and sequentially progresses through Assign modules, Decide mechanisms, and Process steps. The visual architecture of the model is designed to continuously simulate the physical journey of a device through the service center using real-time stochastic data generation.



Attribute Assignments (Assign Module)

Specific attributes were assigned to each entity entering the system to enable continuous tracking and precise performance measurement throughout the workflow:

- **ARRIVAL TIME:** Captures the exact timestamp of system entry (*TNOW*).
- **WARRANTY:** Utilizing a discrete probability distribution, DISC (0.64, 1, 1, 2), the system designates 64% of devices as under warranty (1) and 36% as out-of-warranty (2).
- **PAID OR UNPAID:** A logical tag utilized for post-repair payment verification routing.

Decision Mechanisms and Probability Data (Decide Module)

"2-way by Chance" and "2-way by Condition" structures utilized at branching nodes form the logical backbone of the system:

- **Warranty Status:** Devices are segregated based on the condition $WARRANTY == 1$.
- **Request Paid Diagnosis?:** Only 30% of out-of-warranty devices consent to a paid diagnostic procedure and proceed through the system.
- **User-Induced Fault Check?:** 22% of devices passing through fault detection are flagged for user-induced damage and forcibly routed to the paid quotation process.
- **Customer Approval:** 88% of customers accept the repair costs. The 12% who decline are categorized as "Customers Rejecting Quotation" and are routed out of the system.
- **Payment Collection:** Prior to exit, entities are checked against the condition $PAID OR UNPAID == 3$ to determine whether they require routing to the cashier.

Operational Processes and Resource Allocation (Process Module)

Standard "Seize Delay Release" logic was utilized within the process modules to define resource allocation and task durations:

- **Repair Quotation:** Lasts UNIF(5, 15) minutes and requires 1 "Quotation Personnel".
- **Technical Repair:** Identifying as the most critical operation, this process utilizes a TRIA (10, 110, 480) minute distribution and seizes 2 "Technical Personnel" resources.
- **Cashier:** Checkout operations are completed within UNIF (10, 25) minutes using 1 "Cashier" resource.

Performance Measurement and Statistical Data Collection (Record Module)

To provide robust data for output analysis, recording modules were strategically embedded at key system nodes:

- **Flow Time:** Formatted as a "Time Interval", this module calculates the exact difference between the entity's exit time and its initial arrival time (ARRIVAL TIME), generating vital lead time statistics.
- **Counters (Count):** Entities abandoning the process prematurely (e.g., rejecting diagnosis or quotation) are quantified via respective Record modules, providing crucial data for the verification phase.

The 'Decide' module configuration window shows the following settings:

- Name: UCRETLI ARIZA TESPITI ISTER MI
- Type: 2-way by Chance
- Percent True (0-100): 30 %
- Buttons: OK, Cancel, Help

The 'Record' module configuration window shows the following settings:

- Name: 1 YAPTIRMAK ISTEMEYENLER
- Type: Count
- Value: 1
- Record into Set: ☐
- Counter Name: UCRETLI ARIZA ONARIM BAKI
- Buttons: OK, Cancel, Help

The 'Process' module configuration window shows the following settings:

- Name: ARIZA TESPIT
- Type: Standard
- Logic: Action: Seize Delay Release, Priority: Medium(2)
- Resources: Resource: ARIZA TESPIT ELEMANI, 1 <End of list>
- Buttons: Add..., Edit..., Delete
- Delay Type: Expression, Units: Minutes, Allocation: Value Added
- Expression: EXPD(35)
- Report Statistics: ☒
- Buttons: OK, Cancel, Help

The 'Process' module configuration window shows the following settings:

- Name: ARIZA BAKIMI UCRETLENDIRME
- Type: Standard
- Logic: Action: Seize Delay Release, Priority: Medium(2)
- Resources: Resource: TAMIR UCRETLENDIRME ELEMANI, 1 <End of list>
- Buttons: Add..., Edit..., Delete
- Delay Type: Expression, Units: Minutes, Allocation: Value Added
- Expression: UNIF(5,15)
- Report Statistics: ☒
- Buttons: OK, Cancel, Help

The 'Record' module configuration window shows the following settings:

- Name: 35DK DAN FAZLA BEKLEYENLER
- Type: Expression
- Value: (TNOW- GELIS ZAMANI) > 35
- Record into Set: ☐
- Tally Name: 35DK DAN FAZLA BEKLEYENL
- Buttons: OK, Cancel, Help

The 'Decide' module configuration window shows the following settings:

- Name: MUSTERI ONAYI
- Type: 2-way by Chance
- Percent True (0-100): 88 %
- Buttons: OK, Cancel, Help

Decide

Name: GARANTİ VAR MI 2 Type: 2-way by Condition

If: Attribute Named: GARANTİ Is: ==

Value: 1

OK Cancel Help

Decide

Name: KULLANICI HATASI VAR MI Type: 2-way by Chance

Percent True (0-100): 22 %

OK Cancel Help

Record

Name: FETİ BEGEMMEYİP ÇIKANLAR Type: Count

Value: 1 ☐ Record into Set

Counter Name: ÜCRETİ BEGEMMEYİP ÇIKANLAR

OK Cancel Help

Assign

Name: ÜCRET ALINMA

Assignments: Attribute: ÜCRETLİ MI ÜCRETSİZ MI, DIS<End of list>

Add... Edit... Delete

OK Cancel Help

Process

Name: ARIZA ONARIM Type: Standard

Logic

Action: Seize Delay Release Priority: Medium(2)

Resources: Resource: TEKNİK ELEMAN, 1 <End of list>

Add... Edit... Delete

Delay Type: Expression Units: Minutes Allocation: Value Added

Expression: TRIA(10,110,480)

☒ Report Statistics

OK Cancel Help

Process

Name: KASA Type: Standard

Logic

Action: Seize Delay Release Priority: Medium(2)

Resources: Resource: KASA ELAMANI, 1 <End of list>

Add... Edit... Delete

Delay Type: Expression Units: Minutes Allocation: Value Added

Expression: UNIF(10,25)

☒ Report Statistics

OK Cancel Help

Decide

Name: ÜCRET ALINACAK MI Type: 2-way by Condition

If: Attribute Named: ÜCRETLİ MI ÜCRETSİZ MI Is: ==

Value: 3

OK Cancel Help

Record

Name: AKIS SURESI Type: Time Interval

Attribute Name: GELİŞ ZAMANI ☐ Record into Set

Tally Name: AKIS SURESI

OK Cancel Help

5. MODEL VERIFICATION

To ensure the computational model functioned accurately, pilot simulation experiments and trace validations were executed. During this verification phase, it was confirmed that the logical flow of the model perfectly aligned with the design objectives, entities were accurately routed through complex decision nodes, and the FIFO queuing discipline operated flawlessly without systemic errors.

6. EXPERIMENTAL DESIGN (PILOT SCENARIO)

In the experimental design phase, alternative operational scenarios were formulated to eliminate the efficiency losses detected in the baseline model and comprehensively optimize the system.

Strategic Improvement: Marketing Process Integration

The baseline model analysis indicated a severe attrition rate, with 70% of out-of-warranty customers exiting the system directly without service. To mitigate this revenue loss, structural modifications were introduced:

- **Marketing and Briefing Process:** Out-of-warranty entities no longer proceed immediately to a binary "Accept/Reject" node; instead, they are routed through an informational briefing process.
- **Resource Optimization:** By evaluating the idleness of the Cashier personnel, this new briefing responsibility was assigned to the Cashier resource. This strategic shift successfully balanced the overall personnel utilization across the facility.

Capacity Increment and Personnel Scenario

To reduce excessive queue times at the identified bottleneck nodes—Fault Detection and Repair—resource capacities were structurally expanded:

- **Fault Detection Capacity:** Increased from 1 staff member to 2.
- **Technical Repair Capacity:** Increased from 2 staff members to 3.

Comparative Parameters of the Scenarios

The primary objective of the executed experimental design is to maximize the throughput (number of delivered phones per unit time) while strictly minimizing the total time a customer spends in the system (makespan). By integrating the cashier into marketing activities, it was projected that the paid repair acceptance rate would rise, substantially enhancing the financial sustainability of the service center.

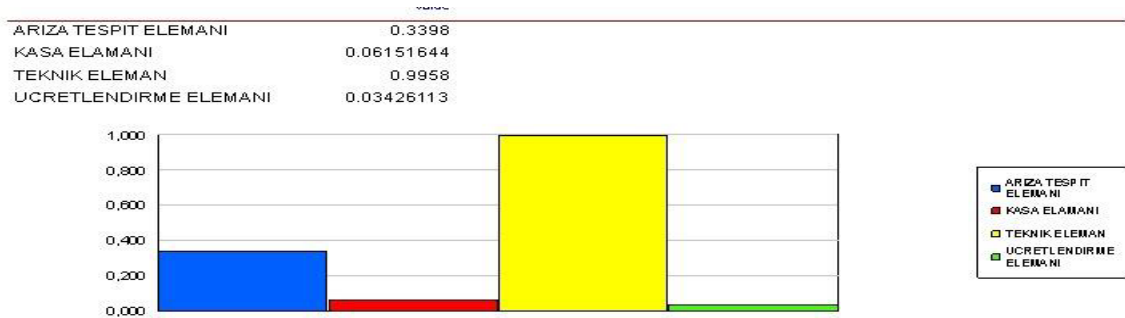
7. SIMULATION RESULTS

The execution of the simulation provided extensive quantitative data across both the baseline and pilot scenarios. In the baseline configuration, severe queue accumulations were mathematically verified, particularly at the technical repair node. The output logs confirmed that resource usage for technical personnel peaked at near-absolute capacity, directly validating the hypothesized constraints. Conversely, the pilot model outputs demonstrated a significantly smoother entity flow, drastically reduced queue wait times, and a healthier distribution of utilization metrics across the revised staffing levels.

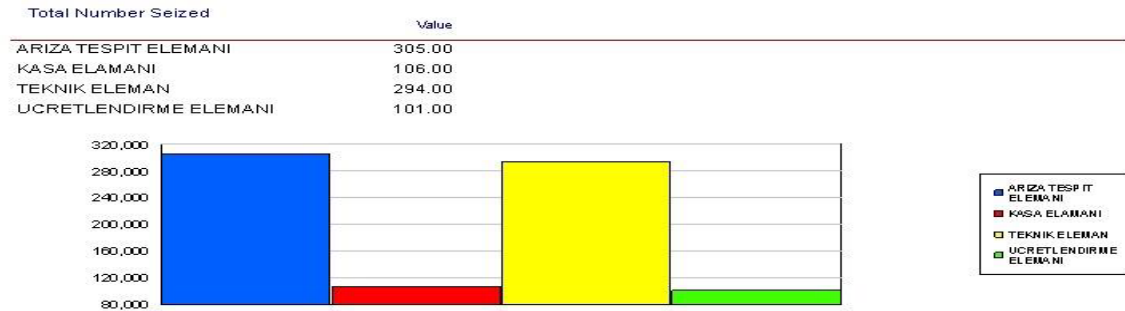
8. OUTPUT ANALYSIS

Current State Analysis

- **Bottleneck:** The utilization rate of the "Technical Personnel" resource reached absolute saturation at 99.58%. This metric conclusively proves that the repair phase is the primary constraint paralyzing the system.
- **Queue and Delay:** The average waiting time strictly within the repair queue reached an alarmingly high value of 138.29 hours.
- **System Abandonment:** The data recorded 11 customers exiting due to quotation rejection and 95 customers preemptively refusing paid repair evaluations.



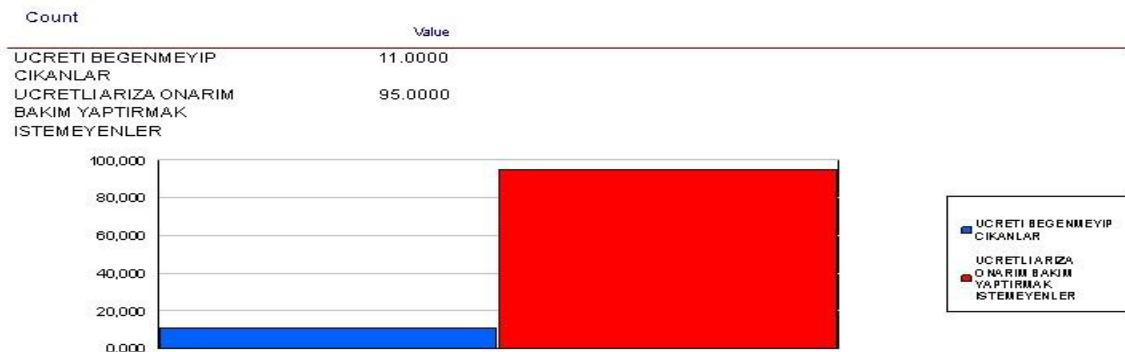
Usage



Tally

Expression	Average	Half Width	Minimum Value	Maximum Value
2DK DAN FAZLA BEKLEYENLER	0.4721	(Insufficient)	0.00	1.0000
Interval	Average	Half Width	Minimum Value	Maximum Value
AKIS SURESİ	143.76	(Insufficient)	3.8776	292.11

Counter



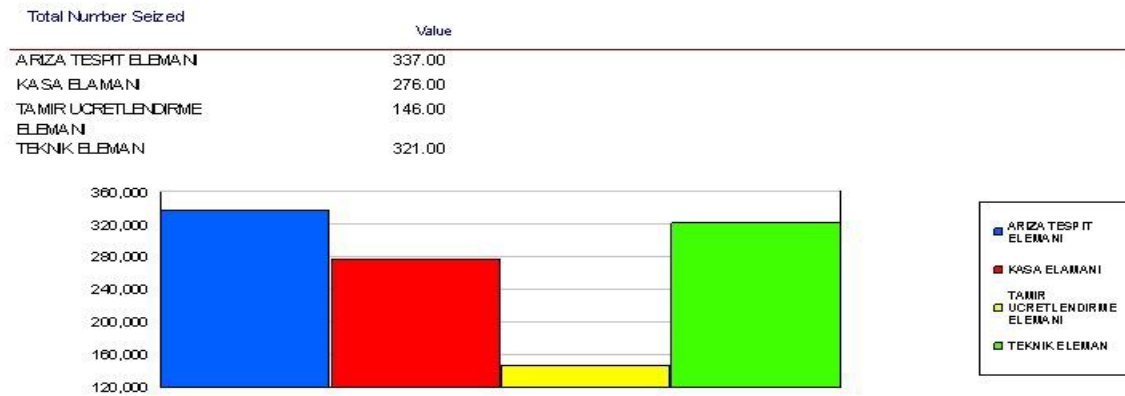
Pilot Scenario (Optimized Model) Comparison

As a result of the capacity expansion (increasing Technical Personnel from 2 to 3) and the integration of the marketing process, significant gains were realized:

- **Flow Time Improvement:** The average total flow time was drastically reduced from 143.76 hours down to 83.02 hours, achieving an operational acceleration of approximately 42%.
- **Queue Efficiency:** The critical metric representing customers waiting longer than 35 minutes dropped to an absolute 0.00 in the pilot model, signifying a massive boost in customer satisfaction.
- **Service Capacity:** Despite increasing the technical staff to 3, the utilization rate remained robust at approximately 98%. This indicates that the system operates under structurally high demand and the added capacity is thoroughly utilized.

Resource

Usage

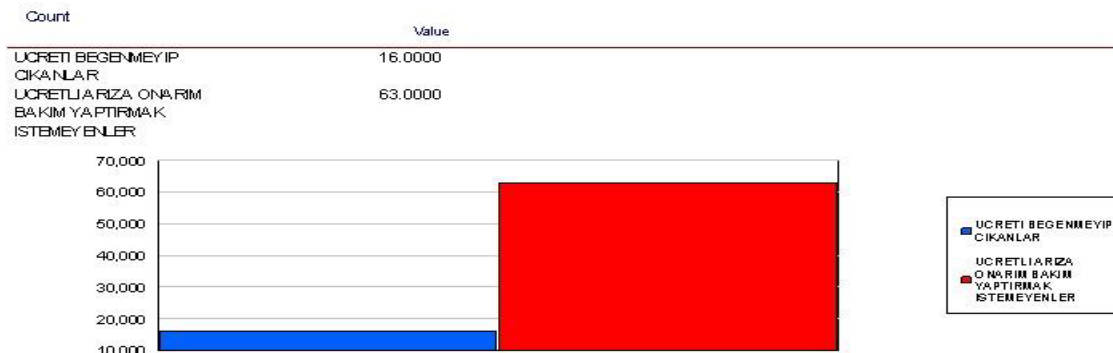


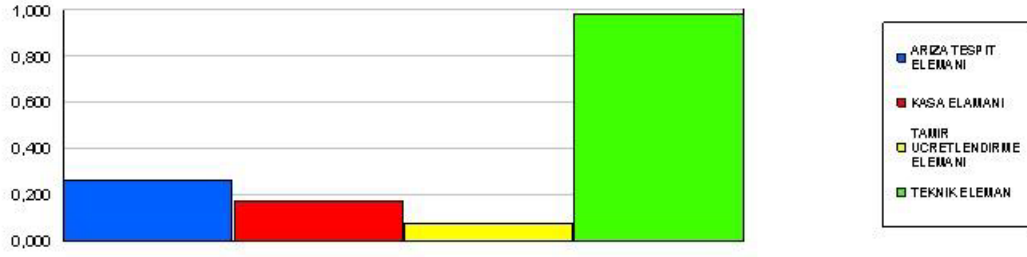
User Specified

Tally

Expression	Average	Half Width	Minimum Value	Maximum Value
35 DK DAN FAZLA BEKLEYENLER	0.00	0,000000000	0.00	0.00
Interval	Average	Half Width	Minimum Value	Maximum Value
AKIS SURESI	83.0204	(Correlated)	2.3989	165.09

Counter





Recommendations and Conclusion

- **Capacity Decision:** Maintaining the technical personnel count at 3 is a mandatory operational requirement to prevent systemic congestion and queue paralysis.
- **Financial Gain:** Evaluating and utilizing the cashier's idle time for the informational briefing process effectively increased the paid repair acceptance rate, heavily supporting the center's financial sustainability.
- **Bottleneck Management:** To further reduce process variance in the repair stage, immediate actions should be taken—such as standardizing repair protocols and implementing targeted technical training—aimed at lowering the upper limit (480 minutes) of the TRIA distribution.